MATH 174 - HW 1

Dani Nguyen

Fall 2025

1. (a) Forward substitution on $Ly = b$

$$\begin{aligned} x_1 &= 2 \\ -x_1 + x_2 &= -1 \\ 3x_1 - 3x_2 + x_3 &= 3 \end{aligned}$$

So, we have

$$\begin{aligned} x_1 &= 2 \\ x_2 &= -1 + x_1 \\ &= -1 + 2 \\ &= 1 \\ x_3 &= 3 - 3x_1 + 3x_2 \\ &= 3 - 3(2) + 3(1) \\ &= 0 \end{aligned}$$

(b) Backward substitution on $Ux = y$

$$\begin{aligned} -1x_1 + 2x_2 + x_3 &= y_1 \\ 2x_2 - 2x_3 &= y_2 \\ 5x_3 &= y_3 \end{aligned}$$

So, we have

$$\begin{aligned}
x_3 &= \frac{y_3}{5} \\
2x_2 &= y_2 + 2x_3 \\
2x_2 &= y_2 + 2\left(\frac{y_3}{5}\right) \\
x_2 &= \frac{1}{2}y_2 + \frac{y_3}{5} \\
-x_1 &= y_1 - 2x_2 - x_3 \\
x_1 &= -y_1 + 2y_2 + \frac{2y_3}{5} + \frac{y_3}{5} \\
&= -y_1 + 2y_2 + \frac{3y_3}{5}
\end{aligned}$$

(c) Compute $A = LU$

$$A = \begin{bmatrix} 1 & 0 & 0 \\ -1 & 1 & 0 \\ 3 & -3 & 1 \end{bmatrix} \begin{bmatrix} -1 & 2 & 3 \\ 0 & 2 & -2 \\ 0 & 0 & 5 \end{bmatrix} = \begin{bmatrix} -1 & 2 & 3 \\ 1 & 0 & -5 \\ -3 & 0 & 20 \end{bmatrix}$$

$$\begin{aligned}
Ax = b: \begin{bmatrix} -1 & 2 & 3 \\ 1 & 0 & -5 \\ -3 & 0 & 20 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} &= \begin{bmatrix} 2 \\ -1 \\ 3 \end{bmatrix} \\
\begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} &= \begin{bmatrix} -1 \\ \frac{1}{2} \\ 0 \end{bmatrix}
\end{aligned}$$

The solution to $Ax + b$ is any multiple of $\begin{bmatrix} -1 \\ \frac{1}{2} \\ 0 \end{bmatrix}$

2. (a) The summation is from $j = 1$ to $i - 1$, there are $i - 1$ additions. There is also one more subtraction between b_i and the summation. So in total, there are i additions and subtractions.

   (b) There are $i - 1$ terms in the summation, therefore there are $i - 1$ multiplications.

3. (a) Gaussian elimination

$$\left[\begin{array}{ccc|c} 2 & -3 & 1 & 1 \\ 1 & 1 & -1 & 2 \\ -4 & 0 & 4 & -1 \end{array}\right]$$

$$\left[\begin{array}{ccc|c} 2 & -3 & 1 & 1 \\ 1 & 1 & -1 & 2 \\ 0 & -6 & 6 & 1 \end{array}\right] R_3 = R_3 + 2R_1$$

$$\left[\begin{array}{ccc|c} 1 & -4 & 2 & -1 \\ 1 & 1 & -1 & 2 \\ 0 & -6 & 6 & 1 \end{array}\right] R_1 = R_1 - R_2$$

$$\left[\begin{array}{ccc|c} 1 & -4 & 2 & -1 \\ 0 & 5 & -3 & 3 \\ 0 & -6 & 6 & 1 \end{array}\right] R_2 = R_2 - R_1$$

$$\left[\begin{array}{ccc|c} 1 & -4 & 2 & -1 \\ 0 & -1 & 3 & 4 \\ 0 & -6 & 6 & 1 \end{array}\right] R_2 = R_2 + R_3$$

$$\left[\begin{array}{ccc|c} 1 & -4 & 2 & -1 \\ 0 & 1 & -3 & -4 \\ 0 & -6 & 6 & 1 \end{array}\right] R_2 = -R_2$$

$$\left[\begin{array}{ccc|c} 1 & -4 & 2 & -1 \\ 0 & 1 & -3 & -4 \\ 0 & 0 & -12 & -23 \end{array}\right] R_3 = R_3 + 6R_2$$

$$\left[\begin{array}{ccc|c} 1 & -4 & 2 & -1 \\ 0 & 1 & -3 & -4 \\ 0 & 0 & 1 & \frac{23}{12} \end{array}\right] R_3 = \frac{-1}{12} R_3$$

Back substitution

$$
\begin{aligned}
x_3 &= \frac{23}{12} \\
x_2 &= 3x_3 - 4 \\
&= \frac{23}{4} - 4 \\
&= \frac{7}{4} \\
x_1 &= 4x_2 - 2x_3 - 1 \\
&= 7 - \frac{23}{6} - 1 \\
&= \frac{13}{6}
\end{aligned}
$$

4.

5.

6.

7.

8.